

Weather & Disease Prediction

PSTAT 134 (Spring 2025) - Final Project: Report

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Introduction

Think that the overlap of weather data and health analytics is in the spotlight right now because of the intended advancement of predictive medicine. Our project would look at this relationship by analyzing a dataset that combines actual weather data with anonymous symptom reports, patient self-reports of their symptoms, and to some extent, the prognosis of illness.

API Usage

We imported our dataset using the kaggle API via kagglehub. More details on the structure of the dataset are below.

Data Description

The data set for this project was obtained from [Kaggle](#) and consists of:

- **Observations:** 5, 200
- **Variables:** 51 Total, 0 Missing
- **Predictors (50 total):**
 - **Numerical:** Age, Temperature ($^{\circ}\text{C}$), Wind Speed ($\frac{\text{km}}{\text{h}}$), Humidity (%)
 - **Binary:** Gender and 45 Medical Symptoms (e.g., Cough, Vomiting, Headache)
- **Response Variable:** Prognosis (e.g., "Arthritis", "Eczema", "Stroke", "Common Cold")

Project Objectives

This project intends to:

- Analyze which of the hydro-meteorological features shape the prognosis of diseases.
- Use clinical symptom and hydro-climate data to accurately predict when people get sick.
- Establish which features contribute the most to accurate prediction.
- Fit prediction models with high sensitivity and precision as measured by the F1 score.

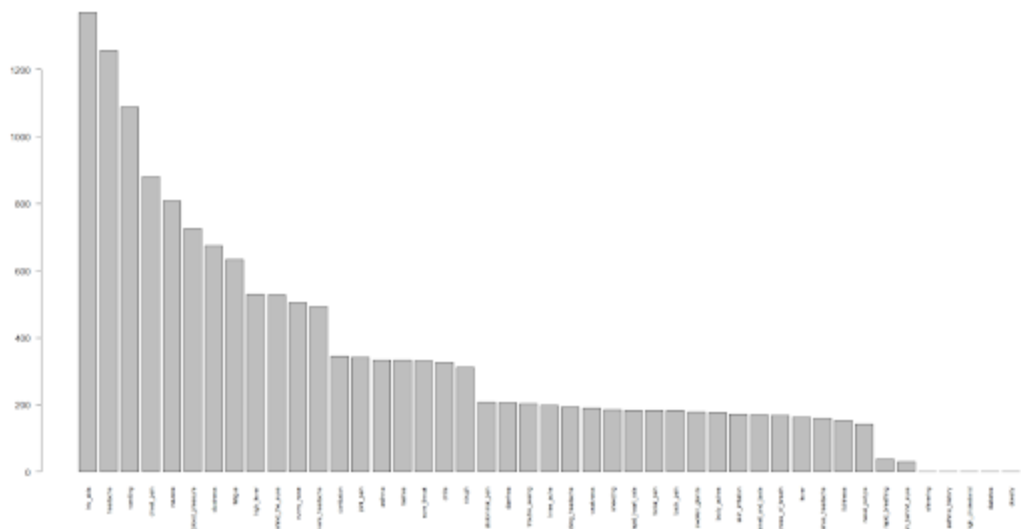
Exploratory Data Analysis

The first stage of exploration showed variation and a balance of symptoms across patients. Additionally, it was noted that certain medical symptoms were more common in specific

prognoses. The weather conditions showed some moderate variation but, again, no clear patterns indicating correlation with disease outcomes at first glance.

Frequency of
Prognosis Categories

Prognosis	Count
Heart Attack	1013
Migraine	941
Influenza	658
Heat Stroke	338
Malaria	330
Stroke	329
Eczema	327
Dengue	322
Common Cold	321
Arthritis	311
Sinusitis	310





Modeling Approaches

Regression-Based Models

We trained several regression-based classifiers:

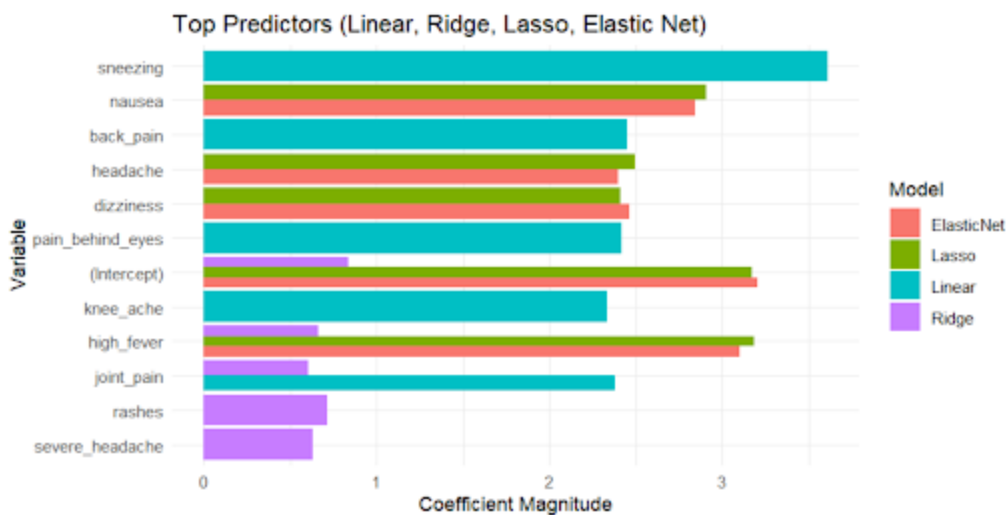
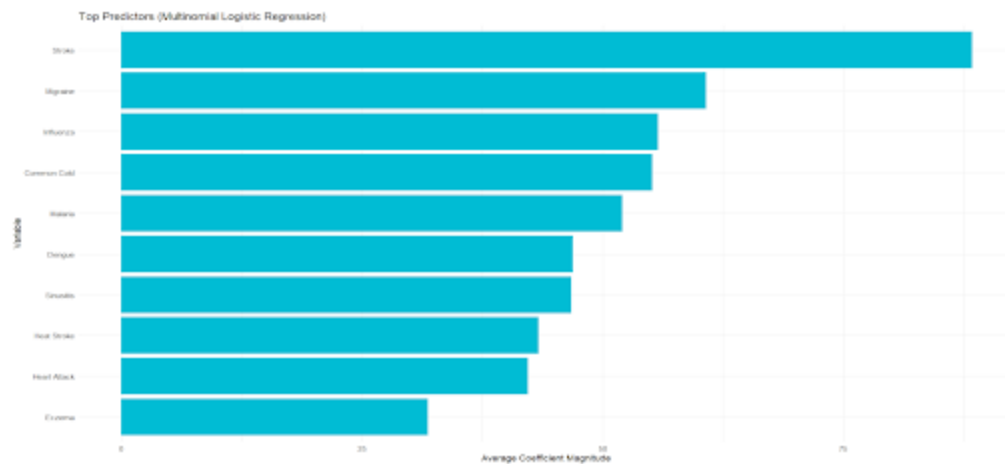
- Linear Regression
- Multinomial Logistic Regression
- Penalized Regression (using α):
 - Ridge Regression ($\alpha = 0$)
 - Elastic Net Regression ($\alpha = \frac{1}{2}$)
 - Lasso ($\alpha = 1$)
- Principal Components Regression

Overall Accuracy: $\sim 83.43\%$ across all models.

Key Predictors:

- **Linear:** *sneezing, back pain, eye pain.*
- **Penalized:** *nausea, headache, dizziness.*

Key Prognoses (Multinomial): *Stroke, Migraine, Influenza*



Neural Networks

We implemented a neural network model to classify patient prognosis. The most accurate model used 3 hidden layers with ReLU activation functions and L2 regularization, then a final layer to output softmax probabilities. It was trained over 100 epochs with early stopping, batch size of 128, Adam optimizer, and the sparse categorical cross-entropy loss function.

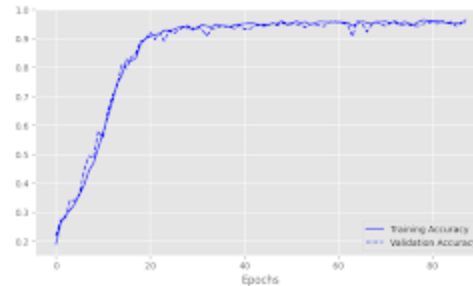
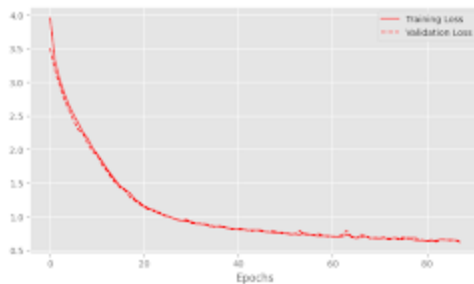
Accuracy: ~ 95.7%

We demonstrate the strong ability of the model to identify complex nonlinear relationships between symptoms and the underlying disease.

```
model = models.Sequential([
    keras.Input((len(feature_selection),)),
    layers.Dense(64, activation='relu', kernel_regularizer=keras.regularizers.l2(0.01)),
    layers.Dense(64, activation='relu', kernel_regularizer=keras.regularizers.l2(0.01)),
    layers.Dense(32, activation='relu', kernel_regularizer=keras.regularizers.l2(0.01)),
    layers.Dense(len(y.unique()), activation='softmax'),
])

model.summary()
model.compile(optimizer=keras.optimizers.Adam(learning_rate=0.001), loss='sparse_categorical_crossentropy', metrics=['accuracy'])
```

Layer (type)	Output Shape	Param #
dense_48 (Dense)	(None, 64)	3,264
dense_49 (Dense)	(None, 64)	4,160
dense_50 (Dense)	(None, 32)	2,080
dense_51 (Dense)	(None, 11)	363



Decision Trees

Three tree-based models:

- Decision Tree
- Random Forest
- Extreme Gradient Boosting (XGBoost)

We applied three tree-based supervised machine learning models --decision tree, random forest, and XGBoost-- in order to predicted the target prognosis using the symptoms and weather features. Using about a 80-20 training and test split, we fit each model on the training data of about 4000 observations. The decision tree model predicted classes by recursively splitting the data based on the most informative features, but was still very complex after tree pruning. To improve model accuracy and reduce possibilities of overfitting, a random forest model was also fit on the training data. This is an ensemble method that takes the results of multiple decision trees trained on bootstrap samples of data with random feature selection to ensure a robust and generalizable model. Furthermore, we trained a XGBoost model, which is another ensemble method that builds decision trees sequentially using gradient boosting and regularization techniques to reduce overfitting and ensure the loss function is minimized.

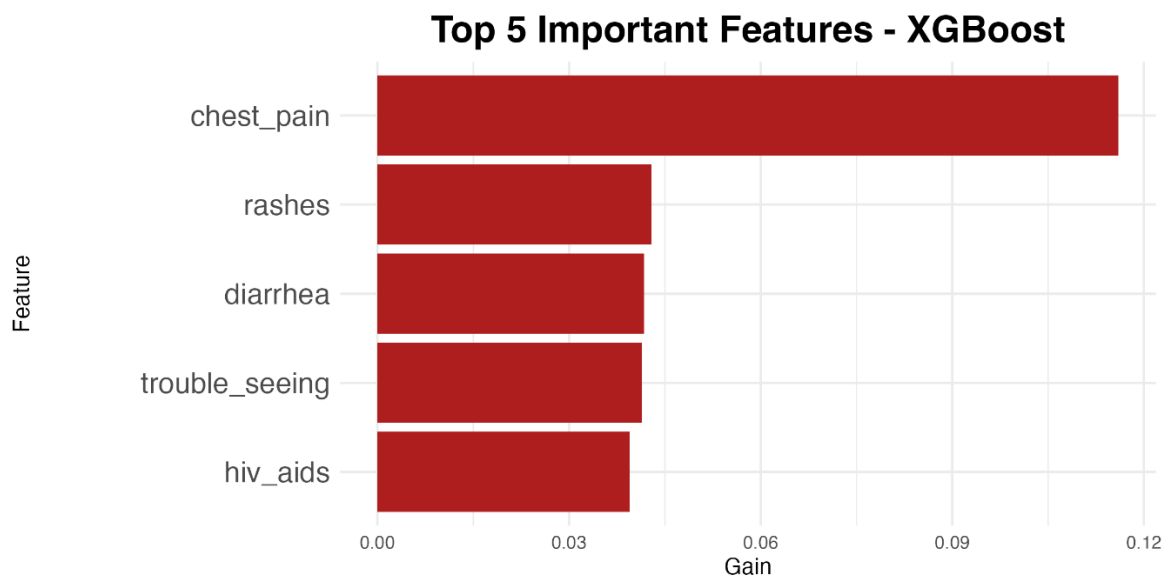
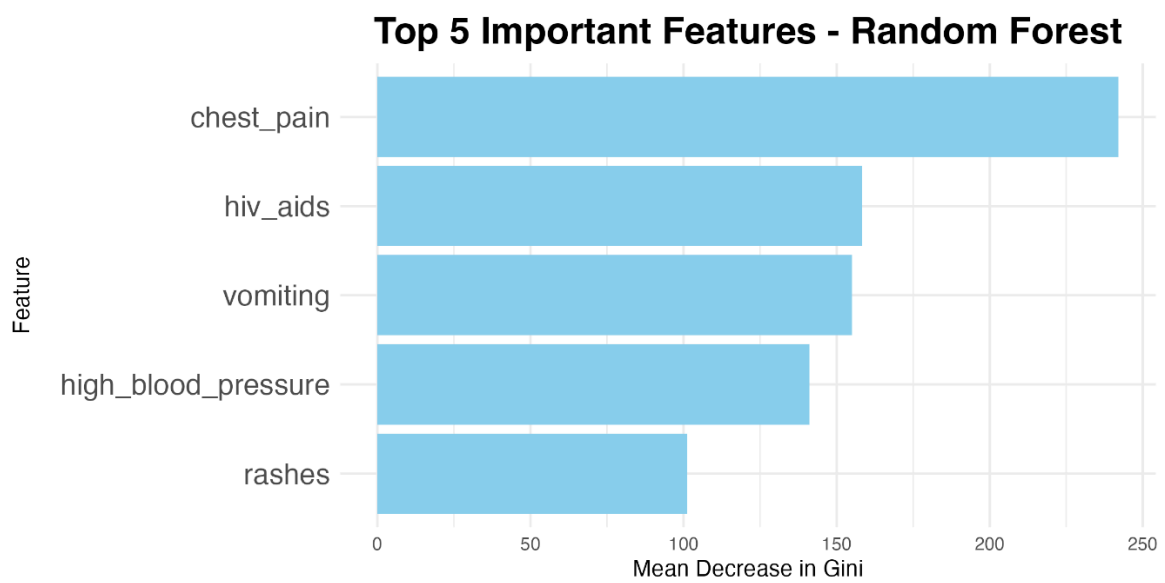
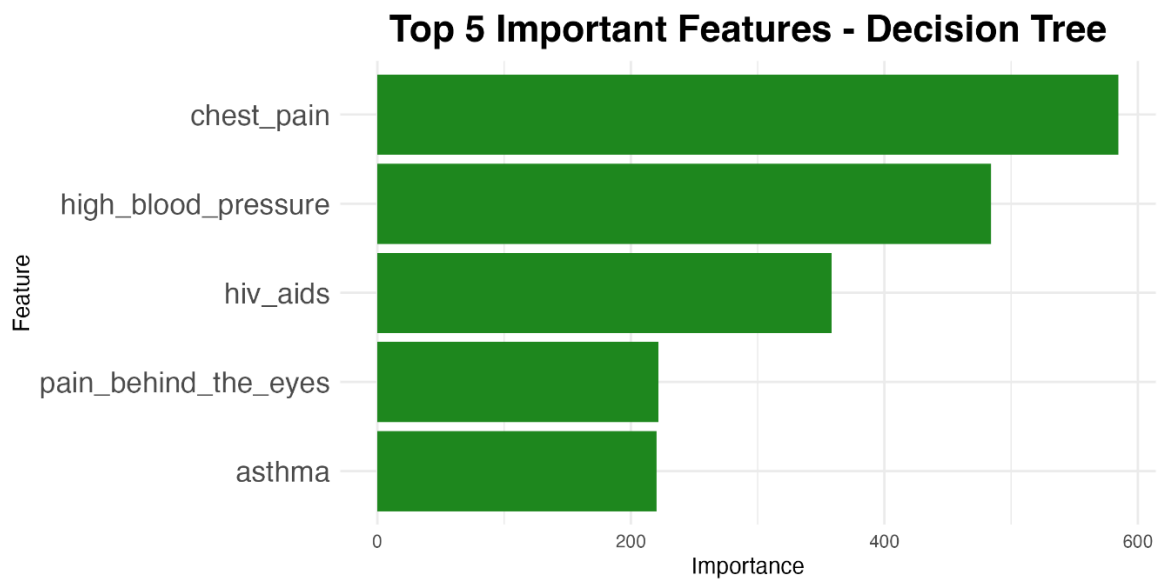
Best Performance: XGBoost, ~ 98.9% Accuracy

Model performance was evaluated using their respective accuracy, precision, recall, and F1-score on the test data. The XGBoost model achieved the highest classification accuracy over all other models at 98.9%, exhibiting its superior robustness and generalizability.

Key Predictors:

- **Decision Tree:** chest pain, high blood pressure, hiv aids.
- **Random Forest:** chest pain, hiv aids, vomiting.
- **XGBoost:** chest pain, rashes, diarrhea

Model Accuracy Comparison				
Model	Accuracy	Precision	Sensitivity	F1
Decision Tree	0.855	0.900	0.821	0.846
Random Forest	0.974	0.980	0.973	0.976
XGBoost	0.989	0.988	0.989	0.988



Results & Conclusions

Best Model: XGBoost ($\sim 98.9\%$ Accuracy)

- **Neural Networks:** Very strong second ($\sim 95.7\%$)
- **Regression Models:** Overall lower accuracy ($\sim 83.43\%$) but revealed the linear relationships.

Weather Features: Surprising not significant in predicting prognosis

Symptom Data: Were driving basically all the model performance. The models demonstrated that symptoms alone are sufficient to classify disease accurately.

Conclusion: Our project shows that machine learning models, especially XGBoost, can classify patient prognosis from symptoms. Weather features, which were thought to contribute to disease, did not make significant contributions in this data set. These findings are encouraging for potential implementations of predictive models in clinical decision support systems to assist in early diagnosis and to instill confidence in patients.

Future Changes: We believe that the 3 weather condition variables in the dataset were insufficient for uncovering the relationship between climate and disease. This relationship can be expanded on by accessing hospitals' anonymized symptoms records, e.g. via VitalDB, and merging them with a dataset of more detailed environmental conditions (precipitation, UV index, biome) retrieved from local weather stations' APIs. We could also add data about population density and demographics, level of access to public services, average income, etc. to further explore the biggest contributing factors to disease.